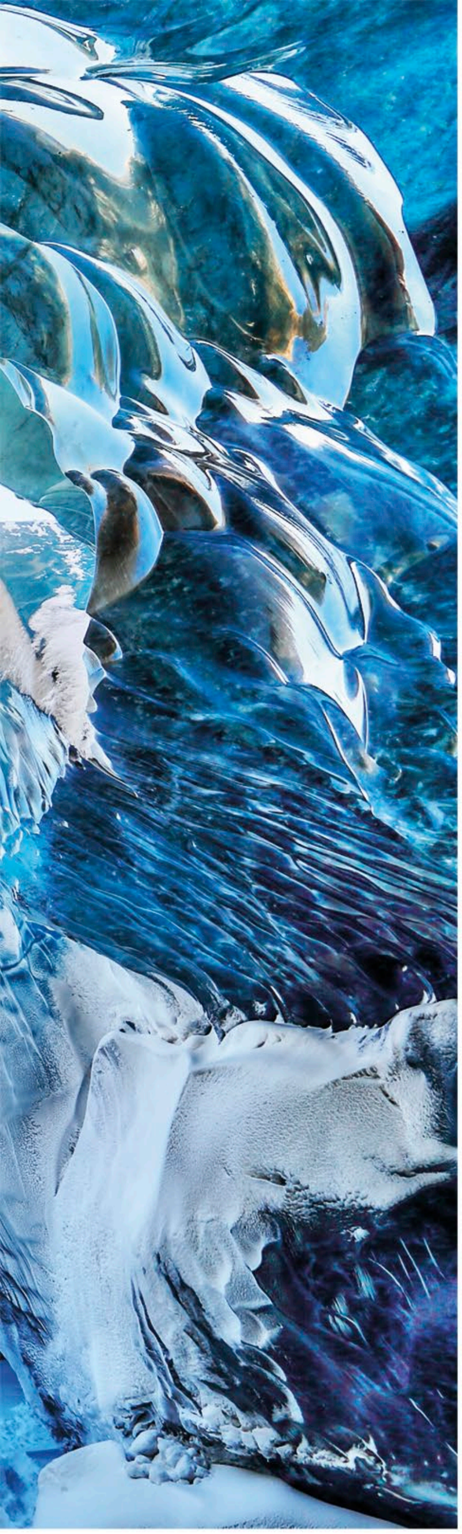


Vatnajökull

A vast ice cap covering 8 percent of Iceland's surface area and the largest glacier by volume in Europe

One of several ice caps in Iceland, Vatnajökull derives its name from the Icelandic words *vatna*, meaning "water," and *jökull*, meaning "glacier." The ice cap completely covers the mountainous terrain of southern Iceland. Under the ice cap are valleys, mountains, and plateaus.

Shape and size

Roughly elliptical, the ice cap forms a frozen dome, with its highest point lying at an altitude of over 6,600 ft (2,000 m). It covers a total area of about 3,130 square miles (8,100 square km). Around 30 outlet glaciers drain the ice from Vatnajökull, carrying it toward the sea. Some of the ice breaks off as icebergs into a large glacial lagoon, Jökulsárlón, and from there makes its way via channels to the sea. Owing to the light-bending effects of atmospheric refraction, the ice cap is said to be visible sometimes from the top of the highest mountain in the Faroe Islands, which are 340 miles (550 km) away. This has been recorded as one of the world's longest sight lines.

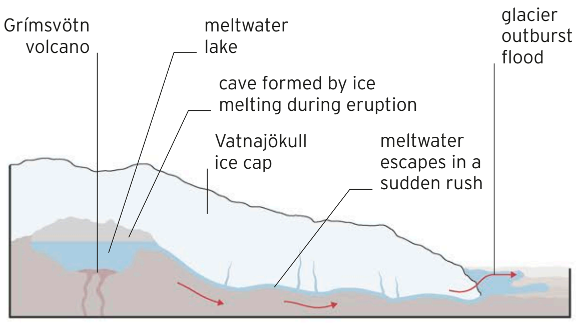
Subglacial volcanoes

Three active volcanoes, as well as some volcanic fissures, lie under Vatnajökull. The active volcanoes are named Bardarbunga and Öræfajökull (both stratovolcanoes), and Grímsvötn, which is a caldera volcano. When one of these volcanoes erupts, along with the usual effects and dangers of a volcanic eruption, there is the added risk of large amounts of ice melting and causing a catastrophic flood (see panel, below). Events of this sort, which are widely referred to by the Icelandic term *jökulhlaups* (meaning "glacier floods"), are most often associated with the Grímsvötn volcano, which has the highest eruption frequency of all Icelandic volcanoes. In November 1996, an eruption of Grímsvötn melted millions of tons of Vatnajökull's ice and triggered a flood that lasted two days. The floodplain between the glacier and the sea became littered with huge blocks of ice.

◁ BORN SWIMMER
One of six seal species found in Iceland, the common, or harbor, seal is often seen in a meltwater lagoon, Jökulsárlón, situated between Vatnajökull and the sea.

FLASH FLOODING

When the Grímsvötn volcano under Vatnajökull erupts, large quantities of the overlying ice melt. This can create such pressure that the ice cap lifts up slightly and vast amounts of water flood out from beneath it. This can devastate the coastal plain next to the glacier.



The **ice cap** has a maximum thickness of **3,300 ft (1,000 m)** and is **1,300 ft (400 m) thick** on average